

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	BS-ARCE
Program Name	Bachelor of Science in Architectural Engineering
Assessment Year	2024-2025
Program Assessment Coordinator	Brent Stephens, Stephen Kleps
Year of Current Assessment Plan	2024-2025
No Assessment This Year per Plan	☐
LO(s) Assessed	6
Presented at program authority meeting date: <i>Please attach minutes</i>	n/a

Methodology

The Bachelor of Science in Architectural Engineering program prepares graduates to enter and be successful in the architectural engineering profession. Graduates are expected to become licensed professional engineers, and to reach responsible positions in a wide range of professional settings, including consulting firms, industry, or government. The program prepares students to begin and successfully complete graduate studies in engineering and/or post-baccalaureate education in a professional degree program. The architectural engineering program provides breadth in core sub-disciplines and depth in areas of specialization. The program is accredited by the Engineering Accreditation Commission of the Accreditation Board for Engineering and Technology (ABET).

Consistent with the program's ABET accreditation, the program has 7 student outcomes (SOs) as learning objectives (LOs) for students completing the program in preparation for entering the professional practice of architectural engineering:

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. an ability to communicate effectively with a range of audiences
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts

5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies

Each year, a detailed assessment report is provided to assess the level of attainment of each student outcome by students in the program, consistent with ABET requirements, which includes both formative and summative assessments (in fact, the majority of the assessment process focused on formative assessments). A full copy of this academic year's report is attached. Here, we focus only on summative assessments to assess the program. To do so, we utilize student performance in a singular course, CAE 495 Capstone Senior Design, which is the culminating design experience for the program, required for all students.

In CAE 495, students are required to integrate multiple engineering disciplines in a group effort to satisfy client requirements for a real engineering project. Students are required to demonstrate mastery in the application of numerous engineering disciplines to a project, work as a member of an integrated engineering team, and demonstrate the ability to understand and communicate engineering solutions to a client verbally, visually, and in written form. The course is annually used to assess student outcomes (i.e., learning objectives) 1-5 and 7 (6 is not covered since it seeks to demonstrate an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions, which we assess only formatively through our earlier laboratory classes).

Here, we report student performance towards these 6 student outcomes (i.e., learning objectives) to assess the program and its graduating students. Each outcome has multiple performance indicators. We average across all performance indicators and all students in the program who took the course to generate a singular assessment of each of the 6 outcomes/objectives. Assessed deliverables include the following:

Table 1. Deliverables used to assess each SO/LO

SO / LO	Deliverables
1	Reports and presentations
2	Reports and presentations
3	Reports and presentations
4	Reports and presentations, Ethics test
5	Digital binder
7	Reports and presentations

Between 2 and 4 specific performance indicators (PIs) are used to assess specific aspects of each SO/LO on a 1-4 score in which 1 = unsatisfactory, 2 = developing, 3 = satisfactory, and 4 = exemplary. Definitions of proficiency levels are as follows: 4 (proficient), 3 (meets expectations), 2 or 1 (does not meet expectations). Scores on individual PIs are averaged to generate a singular score for each of the 6 assessed SOs.

Results

Data

Twenty-one (21) students in the program took CAE 495 in Spring 2025 in this academic year. Figure 1 shows distributions of proficiency levels for SO/LO #1-5 and #7 for this academic year.

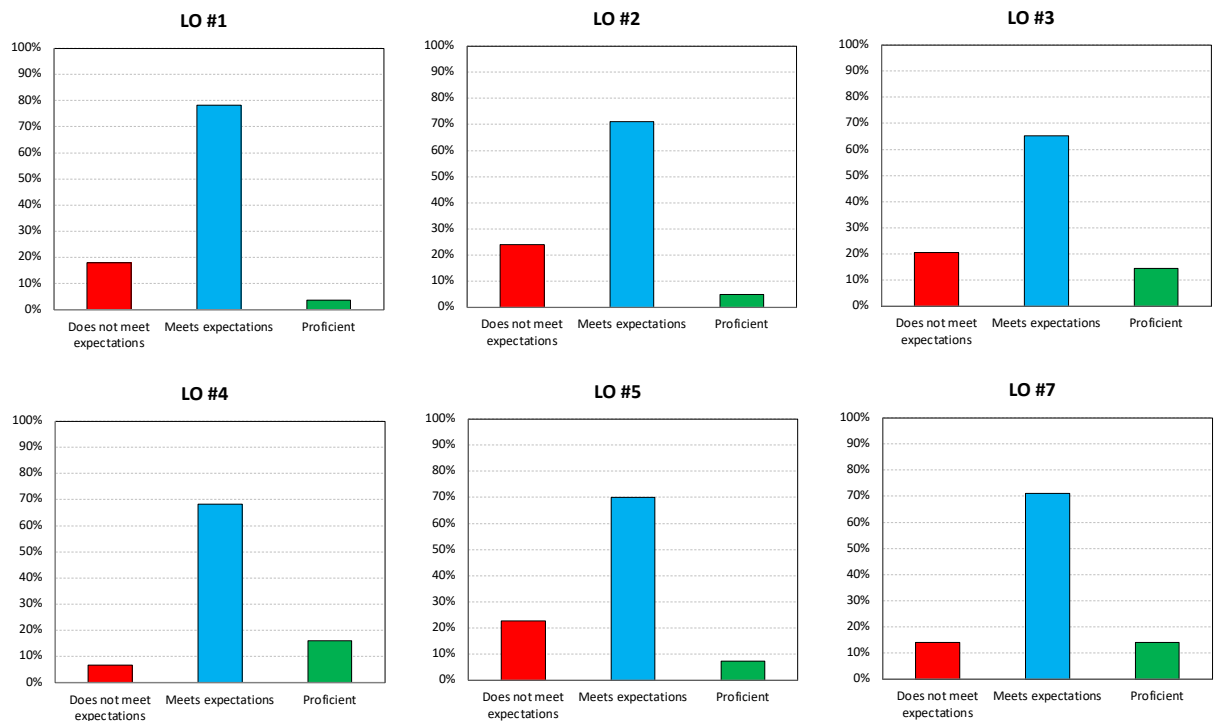


Figure 1. Distribution of proficiency levels for LOs #1-5 and #7 for this academic year.

Description of findings

Key findings from this academic year and cohort include:

- The percentage of students meeting or exceeding expectations exceeded our desired threshold of 70% on all 6 of these assessed summative LOs in this academic year, indicating acceptable progress towards student outcomes and learning objectives by the time of graduation in this cohort.
 - The percentage of students exceeding expectations and demonstrating proficiency ranged from 4% to 16% on these LOs.
 - The percentage of students below expectations ranged from a low of 7% on LO #4 to a high of 24% on LO #2 and 23% on LO #5.
- There were no major concerns flagged in this cycle given this level of performance.

Prior CI plan impact

In previous assessment reports, we had relied solely on our annual assessment and reporting format used to satisfy ABET requirements. In this cycle, we have still conducted that report and included it as attachment herein, but we have also now moved to a simpler reporting format to meet program assessment requirements in a manner that is consistent with the preferred approach and format of the university.

This cycle's CI plan

Please refer to the attachment for more details regarding continuous improvement for the program.

2024-2025 Assessment Report

Bachelor of Science in Architectural Engineering Program

Prepared by:

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October 2025

This annual report covers activities from August 2024 through July 2025.

Program Educational Objectives (PEOs)

The educational objectives of the architectural engineering program are:

- Graduates will meet or exceed the expectations of their employers.
- Graduates will continue to develop professionally by pursuing professional licensure and reach positions of leadership in a wide range of professional settings including consulting firms, industry, academia, or government.
- Graduates will continue to develop professionally by pursuing graduate studies in engineering and/or post-baccalaureate education in a professional degree program.

PEOs were last revised at the May 2020 CAEE Department Advisory Board meeting and presented for annual review by stakeholders at the May 2024 CAEE Department Advisory Board meeting.

Student Outcomes (SOs)

The faculty and its constituents formally adopted on May 1, 2020, at its annual Department Advisory Board meeting the engineering criteria “1” through “7” student outcomes, namely:

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. an ability to communicate effectively with a range of audiences
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies

Assessment of Student Outcomes

The program uses three main approaches to evaluate the level of attainment of student outcomes (SOs) in specific courses and broadly across the curriculum, as shown in Table 1.

Table 1. Student Outcome assessment methods, data type, and frequency

Assessment method	Frequency	Type of Data
1) Instructor course assessments in select required courses	Annually	Assessment of SOs through a series of performance indicators in select courses
2) Fundamentals of Engineering (FE) Examination results	Annually	Passing rates and ratio scores by subject
3) Graduating senior exit survey	Annually	Specific questions on students’ level of preparation for each SO upon graduating from the program

Instructor course assessments are direct assessments of the attainment level of relevant student outcomes within a course using specific assessment items. A set of instructor course assessments in turn provides the data for student outcome assessments. Direct assessments are based on items determined by the instructor, such as quizzes, exam questions, homework problems, presentations, papers, peer reviews and projects. The instructor selects items for assessment according to a rubric of performance indicators (PIs) defined by the department for each student outcome. The course instructor reports the level of performance of each student on each PI for each relevant SO, and the department sets a minimum level of performance that defines expectations. Students who do not achieve at these levels are classified as “below expectations.”

Fundamentals of Engineering (FE) examination results are direct assessments of the attainment level of student outcomes in the program. All students in the program are required to register for the FE exam in their final semester and to show proof of registration prior to graduation. Students are also strongly encouraged to take and pass the exam, reinforced throughout their time in the program. Student performance in terms of pass rates and ratio scores within each subject area is compared to the national average (i.e., ABET comparators). Comparative passing rates provide a broad assessment of program educational outcomes while subject ratio scores provide more specific assessments of student outcomes. Most students in the program take the ‘Other Disciplines’ FE exam.

Graduating senior exit surveys are used to assess several different aspects of the program and the students’ perceived success in attaining student outcomes at the time of graduation. The survey asks students to rate their perceived level of agreement with several questions regarding their level of preparation for beginning their careers, their level of preparation to learn and work independently, their level of preparation for organizing and leadership, the quality of advising received, their perceptions of the quality of faculty inside and outside of the program, the adequacy of facilities and support services, the curriculum (including specific questions on IPRO and senior design experiences), and their level of preparation for the student outcomes in their degree program. Students are asked to rate their responses on a Likert scale of 1 (i.e., disagree or not important) to 5 (i.e., agree or very important). Although generally considered indirect assessment tools, specific survey questions about graduate or professional school plans and membership of student organizations contained in senior exit interview questionnaires are considered “facts” rather than “opinions.” Therefore, they can be used as a direct measure to assess outcomes. Further, feedback from graduating seniors regarding their perceptions of the extent to which they attained the student outcomes can provide valuable information for initiating program improvements.

Assessment planning timeline

Our goal is to conduct each step of the assessment process on an annual basis, including data collection and assessment of SO attainment in individual courses and across courses, FE exam results, and graduating student questionnaires.

Assessment Methods

The primary goal of our assessment of student outcomes (SOs) is to assess whether students in the program have learned what we want them to have learned, with abilities, knowledge, and skills developed over time. To support this goal, we conduct both formative and summative assessments. Instructor assessments of the level of attainment of SOs by students use a standardized set of performance indicators (PIs), common to all specified courses and their instructors. Table 2 shows the standardized PIs that CAEE has adopted for each SO.

Table 2. Standardized performance indicators (PIs) adopted for each Student Outcome (SO)

SO	Description	Performance Indicators (students are able to...):
1	An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics	1) Formulate the problem and identify key issues/variables 2) Identify and apply mathematical and/or scientific methods and equations to solve engineering problems 3) Analyze alternative solutions to an engineering problem 4) Select appropriate techniques and tools for complex engineering problems
2	An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors	1) Produce a clear and unambiguous needs statement in a design project 2) Identify constraints on an engineering design problem and establish criteria for acceptability and desirability of solutions (e.g. economics, environmental, social, political, ethical, etc.) 3) Evaluate and analyze the economics and other relevant factors of an engineering problem solution 4) Incorporate these attributes and constraints in design from planning to construction phase
3	An ability to communicate effectively with a range of audiences	1) Organize material and provide data to support claims or inform audience (e.g., in professional engineering documents such as drawings, construction docs, calculation sheets, lab reports, or project reports) 2) Present content in own words to demonstrate comprehension and target multiple audiences 3) Demonstrate proficient use of English in writing 4) Prepare and deliver an oral presentation
4	An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts	1) Identify the global, economic, environmental, and societal context of an engineering situation 2) Describe ethical and professional responsibilities related to an engineering project 3) Explain the impact of engineering decisions in a global, economic, environmental and societal context
5	An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives	1) Describe participant roles in a team setting and fulfill appropriate roles to ensure team success 2) Accept and integrate input from all team members 3) Communicate among teammates, ask for feedback, and implement suggestions 4) Set goals, establish tasks, monitor team progress, and meet objectives
6	An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions	1) Follow instructions for experiments and required safety rules 2) Develop and conduct an experiment and acquire data on appropriate variables 3) Analyze and interpret experimental data, including the use of statistics to account for experimental error 4) Present results and conclusions within the context of appropriate theoretical models, including explaining observed differences
7	An ability to acquire and apply new knowledge as needed, using appropriate learning strategies	1) Identify and explain necessary techniques, skills, and tools for new situations 2) Apply new techniques, skills and tools to the given situation

Instructor data collection and reporting

Assessments of SOs using standardized PIs must be conducted (and reported) for students in the Architectural Engineering separately from other programs.

Instructors are required to provide the following in their assessment reports:

1. Annually: Records of their assessments of level of SO attainment for *each student* in their assessed course, made using the standardized PIs and rubric (i.e., 1 = unsatisfactory; does not meet expectations, 2 = developing; but does not meet expectations, 3 = satisfactory; meets expectations, 4 = proficient; exemplary). This allows for pooling assessment data across multiple courses, sections, and academic years for more consistent assessment of SO attainment at the program level.
2. Periodically (e.g., every few years): Copies of work products used in assessing attainment of SOs via standardized PIs (including one each of Exemplary, Satisfactory, Developing, and Unsatisfactory, if applicable).

Appendix B expands upon Table 2 and presents a standard rubric to be used for quantifying attainment for each PI and each SO on the abovementioned 1 through 4 scale. Instructors should use these rubrics to assess all PIs for each SO assigned for assessment in their course.

Table 3 shows an example of an SO attainment data collection table to be used by instructors. Instructors are provided with spreadsheet templates pre-populated with the names of each student in the program in their course for collecting these data.

Table 3. Example of SO attainment data collection table to be used by instructors

SO #1 – An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics				
Rubric: 1 = Unsatisfactory; does not meet expectations 2 = Developing; but does not meet expectations 3 = Satisfactory; meets expectations 4 = Proficient; exemplary				
Student	PI 1: Formulate problem and ID variables	PI 2: ID and apply equations and methods	PI 3: Analyze alternative solutions	PI 4: Select tools and techniques
Klaus Daimler	3	3	3	3
Eli Cash	2	3	1	2
Herman Blume	4	4	3	4
Max Fischer	3	3	2	3
...	...	...	...	...

Table 4 shows which courses are used for assessment of attainment of each SO in the program, along with the type of assessment (i.e., summative, formative, or both), and the expected work products to be used in assessment. To the greatest extent possible, instructors are expected to utilize the types of work products recommended in Table 4 for each SO and mapped course, although they have discretion in defining work products to utilize in each offering.

Table 4. Architectural Engineering: Summary of SO assessment in courses and assessment processes

Student Outcome	Courses Assessed	Type	Work Products
1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics	CAE 208 CAE 286 CAE 287 CAE 302 CAE 303 CAE 304 CAE 307 CAE 331 CAE 383 CAE 470 CAE 495	Formative Formative Formative Formative Formative Formative Formative Formative Formative Summative Summative	Formative: HW and exam problems Summative: open-ended design problems
2: An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors	CAE 461* CAE 464* CAE 470* CAE 495	Summative Summative Summative Summative	Summative: design project submissions, with specific articulation of factors considered and assessed
3: An ability to communicate effectively with a range of audiences	CAE 110 CAE 315 CAE 470 CAE 495	Formative Formative Summative Summative	Both: oral and written communications
4: An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts	CAE 110 CAE 470 CAE 495	Formative Summative Summative	Formative: ethics section on final project reports Summative: design project submissions, with specific articulation of contexts considered and assessed
5: An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives	CAE 110 CAE 464 CAE 470 CAE 495	Formative Summative Summative Summative	Formative: first-year analysis and design project submissions Summative: peer- and evaluator-assessments of teamwork on design projects
6: An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions	CAE 105 CAE 307 CAE 315	Formative Formative Formative	Formative: lab reports
7: An ability to acquire and apply new knowledge as needed, using appropriate learning strategies	CAE 110 CAE 470 CAE 495	Formative Summative Summative	Formative: individual and group assignments requiring acquisition of new knowledge Summative: design project submissions

*Note: CAE 461, 464, and 470 were chosen because they are engineering design courses representing the two major areas of synthesis (design-level) in the program: mechanical and construction.

Table 5 shows FE examination subject area results (i.e., ratio scores) from the Other Disciplines FE exam (the exam most commonly taken by students in the program), which are annually mapped to the 1-7 student outcomes.

Table 5. Mapping of Student Outcomes (SOs) to FE exam subjects

Other Disciplines Engineering FE Exam Subject	SO (1-7)
Mathematics	1
Probability and Statistics	1
Chemistry	1
Instrumentation and Controls	6
Engineering Ethics and Societal Impacts	4
Safety, Health, and Environment	1, 7
Engineering Economics	1
Statics	1
Dynamics	1
Strength of Materials	1
Materials	1
Fluid Mechanics	1, 2
Basic Electrical Engineering	1, 2
Thermodynamics and Heat Transfer	1, 2

Assessment Goals

Achievement goals for each assessment tool are described below.

Instructor Course Assessments – For individual courses, when any of the assessments show fewer than 70% of students in the program meeting or exceeding expectations in a particular course, the course and the assessment item is reviewed first by the instructor and associate chair of undergraduate affairs to determine if additional information is necessary to clarify if it was an atypical finding or if it indicates a significant issue and therefore may need addressing immediately or subsequently. If any of the assessment items show fewer than 70% of students in the program meeting or exceeding expectations in a given PI or SO for a given course for two years in a row, the instructor is required to work with the associate chair of undergraduate affairs to develop an action plan to attempt to improve student performance.

FE Exam Results – Student performance is compared with respect to the national comparator average (via ABET comparators) in pass rates and ratio scores within each subject area. In addition to annual performance, a moving average over a 4-year period is used for pass rates and a 6-year average (from last accreditation review cycle to the current cycle) for ratio scores is used for comparison among subject areas. When the annual pass rate or ratio score for IIT students falls more than 10% below the national average (i.e., ratio score of less than 0.9), or when then 4-year average for pass rates or the 6-year average ratio score in a particular subject area falls more than 5% below the national average (i.e., average ratio score of 0.95), the item is flagged for further investigation and potential actions.

Graduating Senior Exit Surveys – Individual items from the graduating senior surveys administered by the CAEE Department have rubrics 1-5 (where 5 = most favorable response and 1 = least favorable response). An item is flagged for evaluation and development of action items to improve the score when the average over a 4-year period falls below 3.5. This criterion of 3.5 is used to establish a target midway between a neutral response and a response of agreement.

Assessment Results

Results of Instructor Course Assessments of SOs

Table 6 provides a list of which of the 1-7 SOs are covered – and importantly, assessed – in required courses in the Architectural Engineering (ARCE) program. Highlighted cells show which SOs will be assessed in which courses, with emphasis on (i) primarily summative assessments of each SO, (ii) formative assessment of some SOs, and (iii) breadth of coverage in the basic areas of the program defined by the ABET program criteria.

Table 6. Mapping of Student Outcome (SO) coverage/assessment in the ARCE program

Course ↓	Student Outcome →	1) Engr./Math/Science	2) Eng. Design & Factors	3) Comm., Range Audiences	4) Ethics & Responsibility	5) Inclusive Teams	6) Exp., Data, Judgment	7) Learning Strategies
CAE 100 Intro to Engineering Drawing and Design		D	D	D	I		D	D
CAE 101 Introduction to AutoCAD Drawing and Design		I	D	D	D		D	D
CAE 110 Professional Practice I		I	D	A	A	A	I	A
CAE 111 Professional Practice II		D	I	I	D	D	I	D
CAE 105 Geodetic Science		D	D	D	D		A	I
CAE 208 Thermal-Fluids Engineering I		A	I		D	I		I
CAE 286 Theory and Concept of Structural Mechanics		A	I		D	I		I
CAE 287 Mechanics of Structural Materials		A	I		D	I		I
CAE 302 Fluid Mechanics		A	I		D	I		I
CAE 303 Structural Design I (Steel)		A	D	I	I	I	D	I
CAE 304 Structural Analysis I		A	D	D	I		D	
CAE 307 Structural Design II (Concrete)		A	A	D	D	I	A	I
CAE 312 Eng. System Analysis (Econ, Prob, and Stats)		D	D	I	I	I	D	I
CAE 315 Materials of Construction		D	D	A	I	D	A	I
CAE 331 Building Science		A	D	I	I		D	D
CAE 383 Electrical and Electronic Circuits		A	D				D	
CAE 461 Plumbing and Fire Protection Design		D	A	I	I	I	D	D
CAE 464 HVAC Systems Design		D	A	I	I	A	D	D
CAE 468 Architectural Design		D	D	D	D		D	D
CAE 470 Construction Methods and Cost Estimating		A	A	A	A	A	D	A
CAE 471 Construction Planning and Scheduling		D	D			I	D	I
CAE 495 Senior Capstone Design		A	A	A	A	A	D	A
CAE 496 Fundamentals of Engineering (FE) Examination		D	D		D		D	D

A = Assessed (i.e. strongly supported)

D = Developed (i.e. supported)

I = Introduced (i.e. minimally supported)

Yellow = Course to be used for assessing attainment of individual SOs

Tables 7 and 8 show results of the instructor assessments of each student outcome and corresponding performance indicators for Fall 2024 and Spring 2025, respectively. Tables 7 and 8 summarize the percentage of students receiving either a 3 (i.e., satisfactory, and meeting expectations) or 4 (i.e., exemplary, and proficient) on each PI within each SO and within each assessed course.

Table 7. Percentage of students meeting or exceeding expectations on 1-7 SOs and PIs in each assessed course: Fall 2024

SO	1	1	1	1	2	2	2	2	3	3	3	3	4	4	4	5	5	5	5	6	6	6	6	7	7
PI→	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	1	2	3	4	1	2	3	4	1	2
Course↓																									
CAE 105																				67	100	89	94		
CAE 110									100	100	100	75	100	100	100	100	100	100	100					100	100
CAE 208	85	77	85	69																					
CAE 286	92	92	92	92																					
CAE 304	100	100	80	80																					
CAE 315									91	91	91	91								82	82	91	91		
CAE 331	70	70	80	80																					
CAE 383	78	100	67	89																					
CAE 461					94	81	88	88																	
CAE 470	89	89	100	100	100	89	89	100	100	100	100	100	100	100	100	100	100	100	100					89	89
Average for Program	86	88	84	85	97	85	88	94	97	97	97	89	100	100	100	100	100	100	100	75	91	90	93	94	94

Table 8. Percentage of students meeting or exceeding expectations on 1-7 SOs and PIs in each assessed course: Spring 2025

SO	1	1	1	1	2	2	2	2	3	3	3	3	4	4	4	5	5	5	5	6	6	6	6	7	7
PI→	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	1	2	3	4	1	2	3	4	1	2
Course↓																									
CAE 287	92	92	92	92																					
CAE 302	53	29	24	29																					
CAE 303	91	82	91	73																					
CAE 307	89	89	89	89																78	78	78	78		
CAE 464					89	89	89	89								56	56	78	78						
CAE 470	75	100	100	100	100	75	100	100	100	100	100	100	100	100	100	100	100	100	100					75	75
CAE 495	76	76	100	76	76	76	76	76	76	76	76	90	100	90	90	76	76	81	76					86	86
Average for Program	79	78	83	77	88	80	88	88	88	88	88	95	100	95	95	78	77	86	85	78	78	78	78	81	81

The average percentage of students in the program meeting or exceeding expectations (i.e., attaining a 3 or 4 on a scale of 1 through 4) on each performance indicator (PI) for each student outcome (SO) from all instructor assessments across all courses was at or above 70% for all PIs in both Fall 2024 and Spring 2025.

This first-year class showed good preparation in formative SOs 3, 4, 5, 6, and 7, assessed via CAE 110 and CAE 105, with one exception of slightly low scores on PI 1 for SO 6 (follow instructions for experiments and required safety rules).

This sophomore class showed good performance overall on SO 1 in CAE 208, 286, and 287, mostly meeting or exceeding expectations above target thresholds, except for all PIs on SO 1 in CAE 302 Fluid Mechanics. Performance on SO 1 in CAE 302 was most notably lowest on PIs 2-4, with a large number of students below expectations.

This junior class showed largely adequate performance in SOs 1, 2, 3, and 6 for most of the assessed courses, with a few exceptions on some individual PIs in some classes.

This senior class demonstrated good summative performance in all courses assessed.

Discussion of Results by Student Outcome

1) An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics

The average percentage of students meeting or exceeding expectations across all courses for SO 1 was above 70% for all PIs in this academic year, albeit with performance levels below this threshold in some PIs in some individual courses (e.g., CAE 208, 302, and 383). The lowest, most concerning scores, were in all PIs for SO1 in CAE 302 Fluid Mechanics; all other lower scores on individual PIs were borderline, just below 70%. While a prior annual assessment report (two years ago) flagged a switch of CAE 302 from junior year to spring of sophomore year for the average student in the curriculum as something to observe for potentially causing problems, last year there were no major concerns, suggesting that the performance is unlikely due to this change in timing. Rather, it may reflect poorer preparation in this class, as well as some extenuating circumstances that affected CAE 302 in Spring 2025. Notably, the instructor suffered a health issue and missed multiple weeks of class, while another faculty member (department chair, Stephens) stepped in to cover the class for a few weeks just before spring break. Additionally, there was a new teaching assistant (TA) for the course who had not served in this capacity before. With the primary instructor out and a new TA, assignments were slower than usual in getting graded and returned. While it is unclear how much these factors played a role in lower scores, they likely attributed at least in part. We will continue to monitor performance in CAE 302 closely moving forward.

In the junior cohort, there were no major concerns, with only one PI in CAE 383 slightly below 70% threshold of meeting or exceeding expectations. This performance is consistent with reasonably strong performance in last year's cycle when the class were sophomores.

The senior cohort demonstrated overall good performance on SO 1, with no low scores below thresholds. A few scores on some PIs in CAE 495 were somewhat borderline, but nothing systematically concerning.

2) An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors

The average percentage of students meeting or exceeding expectations across all courses for SO 2 was above 70% for all PIs in this academic year. Overall, although there were lower percentages for SO 2 in CAE 495, there were no major concerns on this SO in this academic year.

3) An ability to communicate effectively with a range of audiences

The average percentage of students meeting or exceeding expectations across all courses for SO 3 was above 70% for all PIs in this academic year, with no major concerns in any course. The first-year cohort demonstrated good performance on this SO for all PIs in CAE 110, while the senior cohort also did so in CAE 470 and CAE 495. There were no major concerns on this SO in this academic year.

4) An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts

The average percentage of students meeting or exceeding expectations across the two courses assessed for SO 4 in this academic year was above 70% for all PIs in all courses assessed. This success reflects in part the action items from last cycle in which instructors sought to be more explicit in drawing specific attention to these requirements and provide students resources they need to improve, including examples. Performance was overall very strong this cycle.

5) An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives

The average percentage of students meeting or exceeding expectations across all courses for SO 5 was above 70% for all PIs in this academic year. There were two low scores, however, on PIs 1 and 2 in CAE 464 in Spring 2025. Lower scores were attributed to challenges in team member collaboration and coordination in group projects. Since no other junior-level courses assess SO 5, it remains to be seen whether this pattern continues in the next cycle, and will thus be observed closely. Prior deficiencies on this SO in CAE 495 were not apparent this year.

6) An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions

The average percentage of students meeting or exceeding expectations for SO 6 was above 70% for all PIs in this academic year. This SO is assessed via two junior-level courses (CAE 307 and 315) and one first-year course (CAE 105, which is also taken by a mix of students beyond first-year students). CAE 323 was removed from assessment this cycle because it is no longer a required course in the program. The percentage of students meeting or exceeding expectations was slightly below 70% (i.e., 67%) for PI 1 in CAE 105 (following instructions for experiments and required safety rules). There were no major concerns for SO 6 in this cycle.

7) An ability to acquire and apply new knowledge as needed, using appropriate learning strategies

The average percentage of students meeting or exceeding expectations for SO 7 was above 70% for all PIs in this academic year. This includes one formative assessment (entering first-year students in CAE 110) and three summative assessments (CAE 470, twice, and 495). There were no concerns for SO 7 in this cycle.

FE Exam Results

Two types of results are provided in this section: (1) FE exam passing rates and (2) subject area FE exam scores.

Figures 1 and 2 show the annual and 4-year average FE exam passing rates for students in the program versus the national pass rates, dating back to 2008. Table 11 summarizes the last 9 years of FE exam taking/passing data for students in the program, while Table 12 provides the same data from Figures 1 and 2 but in tabular form. Figures 1 and 2 and Tables 11 and 12 aggregate across all area-specific exams taken by students in the program (i.e., both Other Disciplines and Civil), as students can choose which exam they take. For 2024-2025, only 6 students in the program took the FE exam; 5 took the Other Disciplines exam and 1 took the Civil exam.

Tables 13 and 14 show average FE exam scores by FE exam subject area attained by students in the program taking the FE exam, also compared to national averages. Only subject area results from the Other Disciplines exam are shown in these tables since most students took this exam in this academic year. Subject areas with ratio scores less than 0.9 are highlighted in orange color and flagged for further review.

Table 11a. FE Exam Passing Percentages for 2016-2017 (see Footnotes)

Item	Institution	ABET Comparator (*2)
No. Examinees Taking (*1)	4	161
No. Examinees Passing	3	128
Percent Examinees Passing	75%	80%

Table 11b. FE Exam Passing Percentages for 2017-2018 (see Footnotes)

Item	Institution	ABET Comparator (*2)
No. Examinees Taking (*1)	6	273
No. Examinees Passing	5	199
Percent Examinees Passing	83%	73%

Table 11c. FE Exam Passing Percentages for 2018-2019 (see Footnotes)

Item	Institution	ABET Comparator (*2)
No. Examinees Taking (*1)	11	288
No. Examinees Passing	9	215
Percent Examinees Passing	82%	75%

Table 11d. FE Exam Passing Percentages for 2019-2020 (see Footnotes)

Item	Institution	ABET Comparator (*2)
No. Examinees Taking (*1)	5	135
No. Examinees Passing	3	107
Percent Examinees Passing	60%	79%

Table 11e. FE Exam Passing Percentages for 2020-2021 (see Footnotes)

Item	Institution	ABET Comparator (*2)
No. Examinees Taking (*1)	4	264
No. Examinees Passing	3	167
Percent Examinees Passing	75%	63%

Table 11f. FE Exam Passing Percentages for 2021-2022 (see Footnotes)

Item	Institution	ABET Comparator (*2)
No. Examinees Taking (*1)	7	302
No. Examinees Passing	4	177
Percent Examinees Passing	57%	59%

Table 11g. FE Exam Passing Percentages for 2022-2023

Item	Institution	ABET Comparator (*2)
No. Examinees Taking (*1)	7	327
No. Examinees Passing	6	195
Percent Examinees Passing	86%	60%

Table 11h. FE Exam Passing Percentages for 2023-2024

Item	Institution	ABET Comparator (*2)
No. Examinees Taking (*1)	8	407
No. Examinees Passing	3	246
Percent Examinees Passing	38%	60%

Table 11i. FE Exam Passing Percentages for 2024-2025

Item	Institution	ABET Comparator (*2)
No. Examinees Taking (*1)	6	326
No. Examinees Passing	4	199
Percent Examinees Passing	67%	61%

* Footnotes:

- 1) 0 examinees have been removed from this data because they were flagged as a random guesser.
- 2) Comparator includes all examinees from programs accredited by the ABET commission noted.

Percentage of students passing FE examination

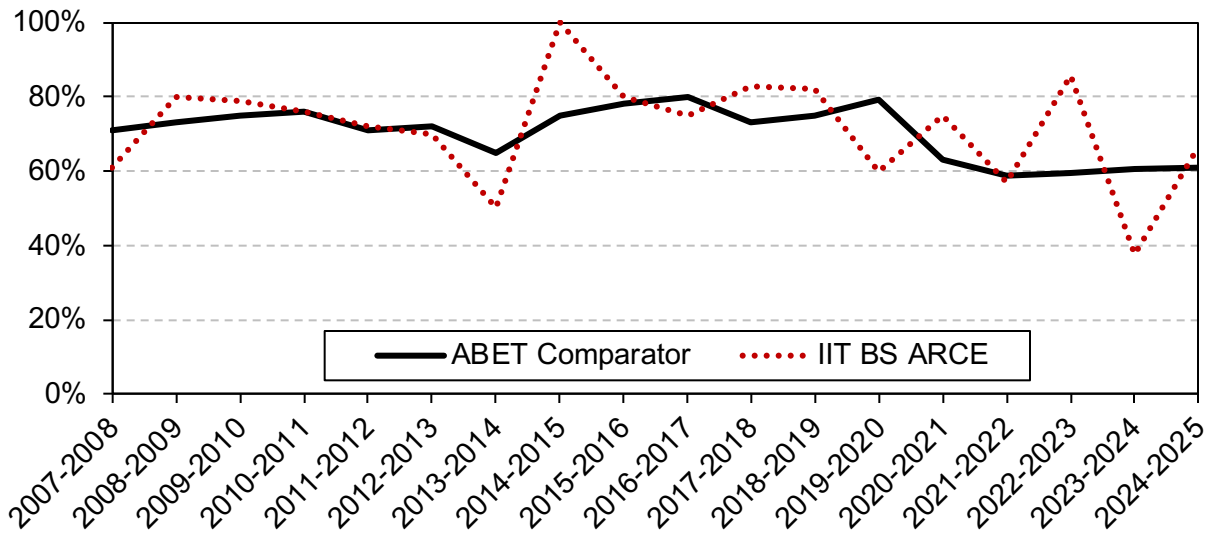


Figure 1. Percentage of students in the program passing the FE examination annually since 2008: IIT ARCE compared to ABET comparators

Percentage of students passing FE examination
(4-year moving average)

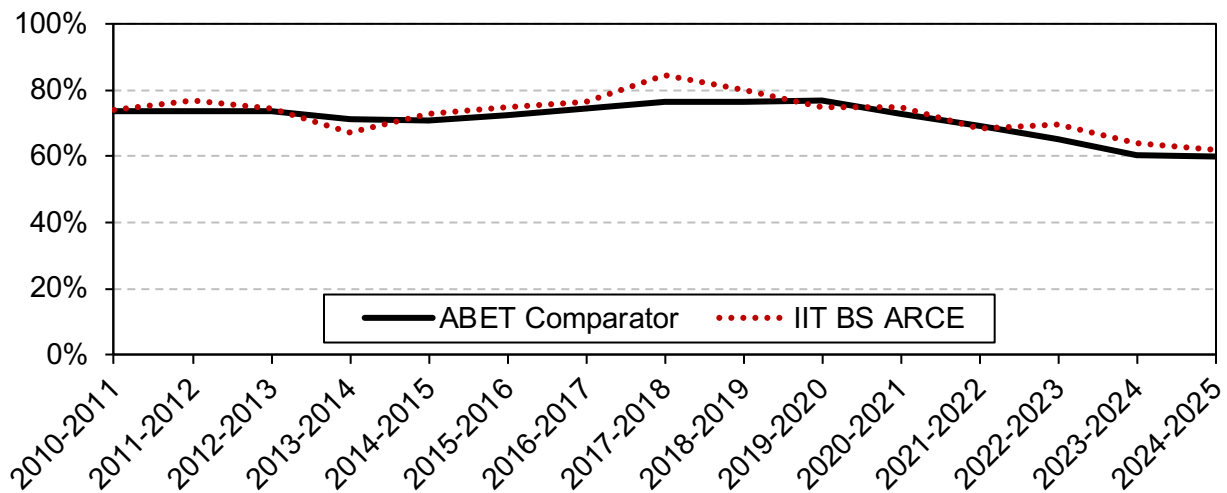


Figure 2. Four-year moving average of the percentage of students in the program passing the FE examination since 2011: IIT ARCE compared to ABET comparators

Table 12. Longitudinal record of FE examination pass rates for students in the program

AY	Annual pass rates			4-year moving average		
	ABET Comparator	IIT	Difference	ABET Comparator	IIT	Difference
2007-08	71%	61%	-10%	-	-	-
2008-09	73%	80%	+7%	-	-	-
2009-10	75%	79%	+4%	-	-	-
2010-11	76%	76%	0%	74%	74%	0%
2011-12	71%	72%	+1%	74%	77%	+3%
2012-13	72%	70%	-2%	74%	74%	+1%
2013-14	65%	50%	-15%	71%	67%	-4%
2014-15	75%	100%	+25%	71%	73%	+2%
2015-16	78%	80%	+2%	73%	75%	+2%
2016-17	80%	75%	-5%	75%	76%	+2%
2017-18	73%	83%	+10%	77%	85%	+8%
2018-19	75%	82%	+7%	77%	80%	+3%
2019-20	79%	60%	-19%	77%	75%	-2%
2020-21	63%	75%	+12%	73%	75%	+2%
2021-22	59%	57%	-2%	69%	69%	0%
2022-23	60%	86%	+26%	65%	69%	+4%
2023-24	60%	38%	-22%	60%	64%	+4%
2024-25	61%	67%	+6%	60%	62%	+2%

In AY 2024-25, only 6 students in the program completed the FE exam and 4 passed (67% pass rate). The 67% pass rate was 6% higher than the ABET comparator average of 61%, reversing a trend from last year in which the pass rate was the lowest single year pass rate in the program in at least 17 years. Our hypothesis was that that cohort was an anomaly, and this cohort suggests that may be the case. The 4-year moving average of the pass rate for students in the program remains at 62%, which is 2% higher than the ABET comparator average of 62%. Our students continue to meet our expectations in this regard, on average, which is hopefully a reflection of the continuous improvement the department has made to its faculty and programming over the last several years.

Of the 5 students who took the Other Disciplines exam, 3 passed (66%), while the 1 student who took the Civil Exam passed. There are no concerns resulting from this year's performance on the FE exam.

Table 13. Results of FE Exam scores for Fall 2024, ARCE taking Other Disciplines (*1)
Examinees taking = 2; examinees passing = 0 (0%; ABET comparators = 57%)

Subjects (*2)	SO	No. of Exam Questions	Institution Average Performance Index (*3)	ABET Comparator Average Performance Index	ABET Comparator Standard Deviation	Ratio Score	Scaled Score (*4)
Mathematics	1	8	8	9	2.1	0.89	-0.48
Probability and Statistics	1	6	6.1	8.7	1.8	0.70	-1.44
Chemistry	1	5	7.2	8.2	2.3	0.88	-0.43
Instrumentation and Controls	6	4	0	8.3	3.3	0	-2.52
Engineering Ethics and Societal Impacts	4	5	6.6	10.3	3.4	0.64	-1.09
Safety, Health, and Environment	1, 7	6	5.4	8.7	2.6	0.62	-1.27
Engineering Economics	1	6	7.1	8.9	2.4	0.8	-0.75
Statics	1	9	7.2	8.9	1.9	0.81	-0.89
Dynamics	1	9	7.5	8.8	1.8	0.85	-0.72
Strength of Materials	1	9	7.1	9.5	1.8	0.75	-1.33
Materials	1	6	7.6	8.8	2	0.86	-0.6
Fluid Mechanics	1, 2	12	6.9	8.7	1.7	0.79	-1.06
Basic Electrical Engineering	1, 2	6	11.7	8.7	2.1	1.34	1.43
Thermodynamics and Heat Transfer	1, 2	9	8.5	9	1.6	0.94	-0.31

* Footnotes:

- 1) 0 examinees have been removed from this data because they were flagged as a random guesser.
- 2) Comparator includes all examinees from programs accredited by the ABET commission noted.
- 3) Performance index is based on a 0-15 scale.
- 4) Uncertainty range for scaled score is +/- 0.71.

Table 14. Results of FE Exam scores for Spring 2025, ARCE taking Other Disciplines (*1)
Examinees taking = 3; examinees passing = 3 (100%; ABET comparators = 56%)

Subjects (*2)	SO	No. of Exam Questions	Institution Average Performance Index (*3)	ABET Comparator Average Performance Index	ABET Comparator Standard Deviation	Ratio Score	Scaled Score (*4)
Mathematics	1	8	9.2	8.8	2.4	1.05	0.17
Probability and Statistics	1	6	9.5	8.7	2.4	1.09	0.33
Chemistry	1	5	10.7	8.5	2.4	1.26	0.92
Instrumentation and Controls	6	4	10.9	8.9	3.1	1.22	0.65
Engineering Ethics and Societal Impacts	4	5	8.8	10.4	3.5	0.85	-0.46
Safety, Health, and Environment	1, 7	6	8.9	9	2.7	0.99	-0.04
Engineering Economics	1	6	8.5	9.3	3	0.91	-0.27
Statics	1	9	8.3	9	2.4	0.92	-0.29
Dynamics	1	9	8.5	8.6	1.9	0.99	-0.05
Strength of Materials	1	9	9.2	9.3	2.1	0.99	-0.05
Materials	1	6	9.3	8.7	2.6	1.07	0.23
Fluid Mechanics	1, 2	12	9	8.6	1.4	1.05	0.29
Basic Electrical Engineering	1, 2	6	8.8	8.7	2.5	1.01	0.04
Thermodynamics and Heat Transfer	1, 2	9	7.7	8.7	1.7	0.89	-0.59

* Footnotes:

- 1) 0 examinees have been removed from this data because they were flagged as a random guesser.
- 2) Comparator includes all examinees from programs accredited by the ABET commission noted.
- 3) Performance index is based on a 0-15 scale.
- 4) Uncertainty range for scaled score is +/- 0.58.

For the 2 students who took the Other Disciplines exam in Fall 2024, neither of whom passed, there were several subjects with ratio scores below 0.9. However, given the small sample size, not much can be ascertained from these data other than these two students were not well prepared for the FE exam. For the 3 students who took the Other Disciplines exam in Spring 2025, all 3 of whom passed, there were only two subjects with ratio scores below 0.9, and not by much. These results each have small sample sizes and likely reflect individual-level variability in preparation for the FE exams, although we will continue to monitor performance over time and will continue to review and revise our attempts at FE exam preparation (i.e., through CAE 496).

Overall, graduates from the program continue to have a 4-year average FE exam pass rate that is similar to or higher than our peers, and we saw a reversal of last year's poor performance, suggesting last year's cohort performance was likely an anomaly.

Graduating Student Exit Surveys

Table 15 summarizes results from the annual graduating student exit survey given to those students who were graduating from the program in either Fall 2024 or Spring 2025, or who were planning to graduate in Summer 2025. The survey uses 7 survey questions to obtain direct student input on their perception of their level of preparation in each SO in their undergraduate program using a 1-5 rating (where 5 = “Strongly Agree” and 1 = “Strongly Disagree”). The sample size for graduates from the architectural engineering program in the 2024-2025 academic year was smaller than in the most recent past year (n = 3), for reasons we do not know, as we continued a practice of including the survey in CAE 495 Capstone Senior Design, a practice that was initiated in 2022-2023.

No averages on student responses regarding their attainment of abilities for any of 7 SOs fell below the 3.5 threshold for acceptability in this academic year. In fact, the average for all 7 SOs was over 4.0, ranging from 4.33 to 5.00, suggesting that graduating student perception of their level of preparation for all SOs was high. No items regarding SO attainment are flagged for additional investigation in this year.

Table 16 shows averages from several additional survey questions assessed via graduating seniors exit questionnaires that are not directly used to assess SOs 1-7 but that provide additional useful information on students’ perception of their educational program, university, and more.

Regarding these additional survey questions, student agreement with most items varied more than in recent years, due especially to the small sample size. While it is difficult to draw useful conclusions from such a small sample size, the respondents were particularly satisfied with their degree acceptance in industry (5.00), their education preparing them for working and learning independently (4.67), their education providing them with effective organization and leadership experience (4.67), and their education preparing them to start their career (4.67). Students reported lower levels of agreement (i.e., below 3.5) with IPRO courses contributing to professional development (3.00), the quality of humanities/social science faculty (3.00), academic support services (2.67), institutional support (2.33), and the quality of math and science faculty (2.00). It is also notable that students reported an overall positive experience with the senior capstone design course (CAE 495) contributing to their professional development (4.33), which is a marked increase compared to the recent past and is likely attributable to the changes made to the course for this academic year. An additional survey was also issued to students in CAE 495, results of which can be found in a separate document. The one student who responded regarding the FE exam preparation class did not think it helped them prepare for the exam (1.00), with a comment about the course not providing enough coverage of Other Discipline exam topics.

As in past years, we will continue to refine the senior design model and will continue to advocate for resources to renovate spaces. We will also introduce an action item to develop more coverage of the Other Disciplines exam in CAE 496.

Table 15. Assessment of students' average level of preparation in each SO (1-7) assessed via graduating seniors exit questionnaires (5 = "Strongly Agree" and 1 = "Strongly Disagree")

Question: Please assess your level of preparation in each of the areas below. These items are specific to program educational outcomes and student outcomes.	2024-2025 (n = 3)
SO 1. I have an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics	4.33
SO 2. I have an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors	4.67
SO 3. I have an ability to communicate effectively with a range of audiences	5.00
SO 4. I have an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts	4.33
SO 5. I have an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives	4.33
SO 6. I have an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions	4.33
SO 7. I have an ability to acquire and apply new knowledge as needed, using appropriate learning strategies	4.67

Table 16. Additional survey questions assessed via graduating seniors exit questionnaires (numbers show average scores using 1-5 rubrics where 5 = "Strongly Agree" and 1 = "Strongly Disagree")

Additional Survey Questions	2024-2025 (n = 3)
Question: Please rate your level of agreement with the following statements involving your senior design experience, IPRO experiences, and more	
The senior-level capstone design course (CAE 495) contributed to my professional development	4.33
The IPRO courses contributed to my professional development	3.00
My education provided me with effective organization and leadership experience	4.67
My education provided me with the ability to learn and work independently	4.67
The courses I took at Illinois Tech prepared me well for the FE exam (n=1)	4.00
The CAE 496 FE Exam Preparation course was helpful in preparing me for the FE exam (n=1)	1.00
Question: Please rate your level of agreement with the following statements regarding your experiences at IIT involving your degree program, faculty, facilities, reputation of the CAEE department, etc.	
The education I received at Illinois Tech prepared me well to start my career	4.67
My degree is well accepted in the engineering industry	5.00
Facilities in Alumni Memorial Hall (e.g., computers, labs, classrooms) are supportive of the educational programs and their objectives	3.67
The program curriculum was well balanced between theory and design courses	4.00
The quality of advising I received at Illinois Tech was sufficient for me to succeed	4.33
The quality of CAEE faculty met my expectations	5.00
The quality of other engineering faculty met my expectations	4.00
The quality of humanities/social science faculty met my expectations.	3.00
The quality of math and science faculty met my expectations	2.00
Institutional support (library, bookstore, etc.) met my expectations	2.33
Academic support services (Registrar's office, Financial Aid office, etc.) met my expectations	2.67

Action Items from AY 2024-2025

The following action items have been introduced in this academic year in response to assessment of outcomes and engagement with program constituents. Refer to past academic year assessment reports for action items from prior years.

Action 1.

Action Taken:	Develop more coverage of the Other Disciplines FE exam in CAE 496.
Basis for Action:	Student feedback and somewhat lower than ideal scores and pass rates on the FE exam by students in the program.
Date:	AY 2025-2026

Action 2.

Action Taken:	Remove CAE 323 from assessment of SO 6.
Basis for Action:	CAE 323 is no longer required but rather students can choose CAE 323 or CAE 466, and most are choosing CAE 466 in the program.
Date:	AY 2024-2025

Action 3.

Action Taken:	Continue to implement our revised senior design (CAE 495) instructional model.
Basis for Action:	We implemented a new CAE 495 model in which all student teams focus on the same local site (United Center redevelopment) and also get matched with local industry mentors; feedback on this model in this cycle was overwhelmingly positive. The most recent instructor, Deb Steimel-Clair, left the university for another position in early summer 2025. Given the crucial role of this course, the department hired another full-time teaching faculty member, Dr. Robert Ryan, immediately upon her departure in summer 2025, who brings an array of experiences in construction engineering and management and will take over as lead instructor of CAE 495.
Date:	AY 2025-2026

Action 4.

Action Taken:	Advocate for space renovations and raise the relatively low perception of math and science faculty and academic support services to leadership in the college.
Basis for Action:	Relatively low survey scores for these items in this academic year (and past academic years).
Date:	AY 2025-2026

Action 5.

Action Taken:	Closely monitor CAE 302 and sophomore cohort as they move into next year.
Basis for Action:	Very low performance on SO 1 in CAE 302 this cycle, possibly attributable to unique circumstances in the course.
Date:	AY 2025-2026

Action 6.

Action Taken:	Undertake departmental strategic planning exercise and revisit program educational objectives.
Basis for Action:	5 years has passed since our last external review; new dean leadership in the college; it is time to revisit our priorities, culture, and initiatives, especially in the context of evolving industry and societal needs and pressures.
Date:	AY 2025-2026

Appendix A: 2025 CAEE Advisory Board Meeting Minutes

Appendix A provides meeting minutes from the 2025 CAEE Board of Advisers meeting, which was held virtually on May 9, 2025, 10 am central time. Meeting Minutes By; Steve Kleps, Ph.D., P.E., Associate Teaching Professor of Civil Engineering and UG Associate Chair for CAEE.

Pre-Meeting Notes:

The Zoom Call allowed attendants in at 9:45 am.

Meeting started at 10:05 am.

Attendance peaked at 40 participants. See Appendix A for the list.

Meeting Notes:

1. Annual Departmental Updates

- a. Brent Stephens started with introductions, review of the agenda, and showing advisory board members and having each introduce themselves.
- b. Board of Advisor's Roster
- c. Departmental Updates
 - i. Remembering those we've lost this year:
 - ii. Sidney Guralnick 1929-2025
 - iii. Burton Lewis 1925-2024
 - iv. Anatol "Andy" Longinow 1933-2025
- d. Personnel Updates University Leadership
 - i. Provost search is underway and at the campus interview stage
 - ii. Dean of Armour College of Engineering. Lou Cattafesta III will start as Dean in August 2025
- e. Personnel Updates:
 - i. Eduardo Corradi promoted to full teaching professor
 - ii. Kahyn Jeon joining as assistant professor in fall 2025
 - iii. 15 full time faculty and 4 staff
 - iv. Many support faculty also there
 - v. New adjunct faculty:
 1. Ajit Naik, PE is a new adjunct on energy performance
 2. Sania Seilabi
 3. Karoline Qasum
 4. Svetlana Taylor
 - vi. Faculty awards:
 1. Matt Gombeda, PCI merit award and a journal award
 2. Ibrahim Osman, Malcom X award, 2025
 3. Deborah Steimel-Clair, Merit award 2024
 4. Brent Stephens, ASCE award
 - vii. Core Research Areas:
 1. Structural Engineering
 2. Transportation Engineering

- 3. Architectural Engineering
 - 4. Environmental Engineering
- f. Student Highlights:
 - i. Ad campaigns:
 - 1. Genna G. highlighted as part of IIT's new marketing campaign
 - ii. Scholarships
 - 1. Four IIT Students won ASCE 2024-2025 scholarship awards
 - 2. Four IIT Students won ASHRAE-IL 2024-2025 scholarship awards
 - 3. Two students won ICRI awards for Chicagoland area
 - 4. More student awards, many students
 - iii. Zoe Lallas, highlighted as an outstanding Ph.D. graduate for her work in concrete research
- 2. CAE 495 Capstone Senior Design Presentations
 - a. Taught by Deb Steimel-Clair and Laurence Rohter
 - b. Spring 2025 recap
 - c. Sample Final Presentations for Civil and Architectural engineering
 - d. Deb presented on:
 - i. Spring semester only class now
 - ii. QAQC lecture and calculation covers
 - iii. Project management
 - iv. Then the 1901 United center complex project, where all teams worked on the same project
 - v. Civil/site teams, environmental, structural, transit,
 - vi. HOBd team, MEP teams, and HVAC teams
 - vii. Now had industry mentors
 - viii. Student feedback has been good, 67% positive response rate
 - ix. The mentorship program was successful
 - x. The challenges are the number of students and number of groups
 - e. Deb asked of the meeting participants:
 - i. How can the board of advisor members help?
 - ii. They can keep their eye out for large projects they could share with us
 - iii. Host construction site tours
 - iv. Example drawing sets could be provided
 - v. Don't have Revit/BIM families that they could share with us
 - vi. How are the companies using AI or not?
 - f. In response Jean-Alix Peralte volunteered to help by sharing info
 - g. Laurence Rohter complimented the two TA's for getting the Revit models in order for capstone.
 - h. The first capstone presentation was Stormwater Solutions by Michael Sansone and Genna Gaitan
 - i. Then Building Structure's - I presented by Ryan Griepentrog and John Gori.

- j. The Bronzeville High Performance Studio Group presented on the high-performance MEP aspects by Angela Priore, Cole Kincart, and Connor Schommer presented on behalf of the group on a shared computer.
3. Guided Discussion
- a. Brent then resumed the guided discussion.
 - i. Strengths and Challenges introduced from an external review of our department from 5 years ago
 - ii. Strong department and culture
 - iii. Recent growth in research
 - iv. On the horizon:
 - v. ABET requirements: Numerical methods, sustainability, risk and resilience
 - vi. AI in the workforce
 - vii. Leverage university-wide initiatives
 - b. Carl started the discussion first. His friend said that in the Alumni newsletter that the CE and EE departments are being closed down? No one ever heard of this. Likely a miscommunication.
 - c. Then Jamshid discussed the CM program in the new Kazakhstan branch and they were very excited about it.
 - d. Then Jean-Alix mentioned that he went to career fairs at UIUC and others and found that many students are in construction management. He didn't understand why.
 - e. Jamshid, answered it by saying that perhaps Professor David Arditi got our program famous and people wanted to come here for CM. The jobs in CM are very lucrative too, so that attracts students.
 - f. Ray Lemming talked to two Indian ambassadors from the new campus on CM teaching here. Ray said that we teach CM in the US differently than other countries.
 - g. Lipika Swarup, adjunct faculty, teaches CM scheduling, she answered.
 - h. This is the MBA for construction folks. Project controls allows you to jump areas, and takes you into management and not just designing.
 - i. Designing felt limiting, and they wanted to move into management.
 - j. Brent said that the freshman surveys also reflect many students interested in CM.
 - k. Brent asked about AI.
 - i. Jean-Alix Peralte used AI on the marketing side, report writing, data mining.
 - ii. Then Laurence said that AI is everywhere.
 - iii. Lindsay Zanders of WSP, said that AI is in some of their tools already
 - iv. Lemming said that stamping drawings have to verify their own work, not use AI
 - v. Alex, AI solutions help you get started
 - vi. Lindsay said that AI is a big risk if not checked.
 - vii. Ray said that companies should check everything with a human.
 - l. John O'Neil expressed his support for study guides and licensure efforts of our department. Because anyone doing public work needs to be licensed.
 - m. Then Brent presented on the departments newish CAE 496 FE Exam Prep Class.
 - n. We are offering the CAE 496 FE Exam Prep course to help them prepare for the FE exam. 51 students registered for it

- o. John O'Neil mentioned the debate of requiring the FE or not.
- p. Brent showed that we exceeded the civil FE exam average, but ARCH was below the FE exam.

Meeting Adjourned at 11:43 am.

Screenshot of Zoom attendees (JPO is John O'Neil):

 John Gori	 Stephen Kleps (Me)
 JPO	 Brent Stephens (Host)
 Laurence IIT	 Gongkang Fu
 Lindsay Zanders - WSP	 Andrew E. (Alumni Hall)
 Lita Sok	 Angela Priore
 Matt Gombeda	 cbalda
 Mehdi Modares	 Charlie Carter, AISC
 Michael Justice	 Charlotte Jackson
 Michael Sansone	 cole kincart
 MingMing	 Connor Schommer
 Mohammad Heidarinejad	 Dave Lampert
 Ophelia's iPhone	 Deborah Steimel-Clair
 Pedro Dulanto	 dgrabowski
 Roger's Notetaker (Otter.ai)	 Diagnostics's iPhone
 Ryan Griepentrog	 Edoarda Corradi
 Saman Haratian	 Genna Gaitan
 Sanaz Kazemzadehazad	 Jamshid Mohammadi
 Seyedmohammad Miralinaghi	 Jean-Alix Peralte
 Soliman Khudeira	
 You Zhou	
 Zhong	
 Zongzhi Li	

Appendix B: Rubric for Performance Indicators for each Student Outcome (1-7)

Table B-1. Rubric for Performance Indicators for Student Outcome (SO) #1

SO #1 – An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics				
Performance Indicator	1 = Unsatisfactory	2 = Developing	3 = Satisfactory	4 = Exemplary
1) Formulate the problem and identify key issues/variables	- Missing problem formulation - Missing most key issues/variables - Missing most criteria/constraints - Missing most assumptions	- Weak problem formulation - Some issues/variables identified, but many missing - Many constraints missing - Many assumptions missing	- Adequate problem formulation - Most key issues/variables are identified - Almost all criteria presented for ranking alternatives - Almost all constraints identified - Almost all assumptions identified	- Complete and succinct problem formulation - Key issues/variables identified - All relevant criteria presented for ranking alternatives - All relevant constraints justified - All relevant assumptions identified
2) Identify and apply mathematical and/or scientific methods and equations to solve engineering problems	- Mostly incorrect methods or equations used - Little analysis provided - Severely flawed analysis	- Partially incorrect methods or equations used - Partially flawed analysis	- Appropriate analysis approach used - Mostly correct analysis procedure - Mostly correct analysis results	- Well thought out or clever analysis approach - Complete and correct approach and results
3) Analyze alternative solutions to an engineering problem	- Alternative solutions are not presented - Criteria not evaluated - Constraints ignored	- Limited analysis of alternative solutions - Only some criteria not evaluated - Only some constraints considered	- Criteria evaluated with minor errors - Constraints considered with minor errors - Variety of trade-offs are presented in alternative solutions	- Complete evaluation of criteria - Complete evaluation of constraints - All significant trade-offs are presented in alternative solutions
4) Select appropriate techniques and tools for a complex engineering problem	- Mostly incorrect use of tools and techniques - No understanding of limitations of tools and techniques	- Some incorrect use of tools and techniques - Limited understanding of limitations of tools and techniques	- Mostly correct use of tools and techniques - Mostly correct understanding of limitations of tools and techniques	- Correct use of tools and techniques - Full understanding of limitations of tools and techniques

Table B-2. Rubric for Performance Indicators for Student Outcome (SO) #2

SO #2 – An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors				
Performance Indicator	1 = Unsatisfactory	2 = Developing	3 = Satisfactory	4 = Exemplary
1) Produce a clear and unambiguous needs statement in a design problem	Unable to formulate a needs statement for a design problem	Partial formulation of a needs statement for a design problem	Mostly complete formulation of a needs statement for a design problem	Complete formulation of a needs statement for a design problem
2) Identify constraints on an engineering design problem and establish criteria for acceptability and desirability of solutions (e.g. economics, environmental, social, political, ethical, etc.)	- Does not understand the concept of constraints - Does not establish criteria for acceptability and desirability of solutions	- Partial understanding of the concept of constraints - Partial establishment of criteria for acceptability and desirability of solutions	- Mostly complete understanding of the concept of constraints - Mostly complete establishment of criteria for acceptability and desirability of solutions	- Complete understanding of the concept of constraints - Complete establishment of criteria for acceptability and desirability of solutions
3) Evaluate and analyze the economics and other relevant factors of an engineering problem solution	Does not evaluate or analyze the economics and other relevant factors of an engineering problem solution	Limited analysis of economics and other relevant factors	Mostly complete analysis of economics and other relevant factors	Complete analysis of economics and other relevant factors
4) Incorporate these attributes and constraints in design from planning to construction phase	Does not incorporate constraints in engineering design from planning to construction	Limited incorporation of constraints in engineering design from planning to construction	Mostly correct incorporation of constraints in engineering design from planning to construction	Correct incorporation of constraints in engineering design from planning to construction

Table B-3. Rubric for Performance Indicators for Student Outcome (SO) #3

SO #3 – An ability to communicate effectively with a range of audiences				
Performance Indicator	1 = Unsatisfactory	2 = Developing	3 = Satisfactory	4 = Exemplary
1) Organize material and provide data to support claims or inform audience (e.g., in professional engineering documents such as drawings, construction docs, calculation sheets, lab reports, or project reports)	- Little structure or organization - Missing major sections (e.g. problem statement, conclusion, summary, references) - Ideas not expressed clearly nor supported by details	- Confusing structure or organization - Weak major sections (e.g. problem statement, conclusion, summary, references) - Ideas not expressed clearly or details are weak	- Mostly logical and complete structure or organization - Adequate major sections (e.g., problem statement, conclusion, summary, references) - Ideas are generally expressed clearly and details are adequate	- Excellent structure or organization - Well-state major sections (e.g., problem statement, conclusion, summary, references) - Ideas are well-developed and expressed clearly with many details
2) Present content in own words to demonstrate comprehension and target multiple audiences	- Lacking information or information is inaccurate or irrelevant - Significant plagiarism - Presents little understanding of the topic - Cannot be understood by a range of audiences	- Some basic information, but some inaccurate or irrelevant - Some text has been plagiarized - Presents basic understanding of some parts of the topic - Partially accessible by a range of audiences	- Adequate information with a few minor errors or omissions - Text is mostly author's own words - Presents general understanding of the topic - Mostly accessible by a range of audiences	- Exceptional information (accurate and relevant) - Careful and thorough research - All text is author's own - Presents in-depth understanding - Easily accessible by a range of audiences
3) Demonstrate proficient use of English in writing	- Numerous errors in grammar, punctuation, spelling - Many sentences have awkward construction - Does not appear to have been proofread	- Several errors in grammar, punctuation, spelling - Several sentences have awkward construction - Proofreading appears to have been done hastily	- A few errors in grammar, punctuation, spelling - Sentences are mostly well constructed - Needs only minor revisions in proofreading	- Minimal errors in grammar, punctuation, spelling - Varied and creative sentence structure - Demonstrates thorough proofreading and revision
4) Prepare and deliver an oral presentation	- Minimal control of speaking tone, clarity, or volume - Speaker is visibly nervous or does not convey interest - Speaker does not make eye contact - No awareness of physical gestures or facial expression	- Delivery is halting; clarify of speech uneven - Speaker appears disengaged or nervous - Limited eye contact with the audience - Limited awareness of gestures or expressions	- Good speaking voice; recovers from errors - Speaker appears engaged but slightly nervous - Good eye contact with the audience - Use of gestures or expressions, but appears forced at times	- Strong, clear speaking voice - Speaker conveys confidence in talking points - Excellent eye contact with the audience - Use of gestures or expressions conveys energy and enthusiasm

Table B-4. Rubric for Performance Indicators for Student Outcome (SO) #4

SO #4 – An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts				
Performance Indicator	1 = Unsatisfactory	2 = Developing	3 = Satisfactory	4 = Exemplary
1) Identify the global, economic, environmental, and societal context of an engineering situation	• Unable to identify relevant contexts of the problem • Relevant contexts described in an extremely limited fashion	• One relevant context of the four listed context types identified • The one relevant context is described in a rudimentary fashion	• Two or three relevant contexts of the four listed context types identified • At least two contexts described substantively	• Three or four relevant contexts of the four listed context types identified • At least three contexts described thoroughly
2) Describe ethical and professional responsibilities related to an engineering project	• Description of ethical and professional responsibilities absent or extremely limited	• Description of ethical and professional responsibilities is limited	• Description of ethical and professional responsibilities is substantive	• Description of ethical and professional responsibilities is complete and thorough
3) Explain the impact of engineering decisions in a global, economic, environmental and societal context	• Explanation of relevant impacts of engineering decisions absent or extremely limited	• Explanation of engineering decision impacts touch on only one context • Explanation of relevant impacts of engineering decisions is rudimentary	• Explanation of engineering decision impacts touch on two or three contexts • Explanation of relevant impacts of engineering decisions is substantive in the majority of contexts	• Explanation of engineering decision impacts touch on three or four contexts • Explanation of relevant impacts of engineering decisions is at least substantive in all contexts and thorough in the majority of contexts

Table B-5. Rubric for Performance Indicators for Student Outcome (SO) #5

SO #5 – An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives				
Performance Indicator	1 = Unsatisfactory	2 = Developing	3 = Satisfactory	4 = Exemplary
1) Describe participant roles in a team setting and fulfill appropriate roles to ensure team success	- Does not recognize the importance of participant roles for team success - Does not contribute or fulfill assigned or assumed roles	- Some recognition of the importance of participant roles for team success - Some contributions or fulfillment of assigned or assumed roles	- Good recognition of the importance of participant roles for team success - Contributes or fulfills most of their assigned or assumed roles	- Thorough recognition of the importance of participant roles for team success - Contributes or fulfills all of their assigned or assumed roles
2) Accept and integrate input from all team members	Fails to accept or integrate input from team members	Accepts or integrates some input from some team members	Accepts or integrates most input from most team members	Accepts or integrates all input from all team members
3) Communicate among teammates, ask for feedback, and implement suggestions	- Fails to communicate among teammates - Does not ask for feedback - Does not consider or implement suggestions from team members	- Communicates some among some teammates - Seeks some feedback - Considers or implements minimal suggestions from team members	- Communicates among most teammates - Seeks feedback among most team members - Considers or implements suggestions from team members	- Communicates among all teammates - Regularly seeks feedback among all team members - Considers and implements suggestions from all team members
4) Set goals, establish tasks, monitor team progress, and meet objectives	- Does not set goals - Does not establish tasks - Does not contribute to monitoring team progress - Does not contribute to meeting objectives	- Sets goals, but unrealistically or unproductively - Some establishment of tasks - Some contribution to monitoring team progress - Some contribution to meeting objectives	- Sets goals, mostly realistic or attainable - Establishes mostly productive tasks aligned with project goals - Sometimes contributes to monitoring team progress - Usually contributes to meeting objectives with minimal error	- Sets realistic and productive goals - Establishes productive tasks aligned with project goals - Regularly contributes to monitoring team progress - Regularly contributes to meeting objectives

Table B-6. Rubric for Performance Indicators for Student Outcome (SO) #6

SO #6 – An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions				
Performance Indicator	1 = Unsatisfactory	2 = Developing	3 = Satisfactory	4 = Exemplary
1) Follow instructions for experiments and required safety rules	- Does not follow instructions for experiments - Does not follow required safety rules	- Follows instructions for experiments, with some deviations - Follows required safety rules, with some deviations	- Follows instructions for experiments, with minimal deviations - Follows required safety rules, with minimal deviations	- Follows instructions for experiments thoroughly - Follows required safety rules thoroughly
2) Develop and conduct an experiment and acquire data on appropriate variables	- Missing experimental plan - Missing identification of key variables - Missing data collection procedure - Missing data or data collection has significant errors - Collected data are not relevant	- Flawed experimental plan - Missing identification of some key variables - Data collection procedure is poorly formulated - Data acquisition lacks details - Collected data lack relevance	- Adequate experimental plan - Identification of most key variables - Data collection procedure is formulated adequately - Data acquisition includes most details - Most collected data are relevant	- Detailed and thorough experimental plan - Identification of all key variables - Data collection procedure is thoroughly formulated - Data acquisition includes all relevant details - All collected data are relevant
3) Analyze and interpret experimental data, including the use of statistics to account for experimental error	- No analysis or interpretation of data - No inclusion of statistics	- Weak analysis or interpretation of data - Weak inclusion of statistics	- Adequate analysis and interpretation of data, with some errors - Adequate inclusion of statistics, with some errors	- Detailed and thorough analysis and interpretation of data - Detailed and thorough inclusion of statistics
4) Present results and conclusions within the context of appropriate theoretical models, including explaining observed differences	- No comparison made to theoretical models or comparison made to nonsensical models - Differences are not identified or incorrectly explained - Conclusions are not justified	- Weak comparison made to theoretical models - Comparison made to models miss some important relationships among key variables - Some differences are identified correctly, but poorly explained - Conclusions are weakly justified	- Adequate comparison made to appropriate models - Comparison includes important relationships among most key variables - Major differences are identified correctly, with only minor errors - Conclusions are mostly justified	- Thorough comparison made to appropriate models - Comparison includes important relationships among all key variables - All differences are identified correctly and weaknesses identified - Conclusions are fully justified

Table B-7. Rubric for Performance Indicators for Student Outcome (SO) #7

SO #7 – An ability to acquire and apply new knowledge as needed, using appropriate learning strategies				
Performance Indicator	1 = Unsatisfactory	2 = Developing	3 = Satisfactory	4 = Exemplary
1) Identify and explain necessary techniques, skills, and tools for a new situation	- Identifies only a subset of necessary techniques, skills, and tools - Identifies unrelated techniques, skills, and tools	- Identifies some necessary techniques, skills, and tools - Identifies some unrelated techniques, skills, and tools	- Identifies almost all necessary techniques, skills, and tools - Missing some minor techniques, skills, and tools	- Identifies all necessary techniques, skills, and tools - Does not include unrelated techniques, skills, and tools
2) Acquire and apply new techniques, skills and tools to the given situation	- Applies only a subset of necessary techniques, skills, and tools - Incorrectly applies techniques, skills, and tools	- Correctly applies some techniques, skills, and tools, but missing important items - Incorrectly applies some techniques, skills, and tools	- Correctly applies almost all techniques, skills, and tools - Incorrectly applies only some minor techniques, skills, and tools	- Correctly applies all relevant techniques, skills, and tools - Does not apply unnecessary techniques, skills, and tools